Analytical Report: Uncovering Market Dimensions in the Brazilian Automotive Industry

Charles Yamamura, March 31, 2026

Objective

The primary objective of this analysis is to simplify a complex dataset of vehicle attributes to identify the latent dimensions that drive consumer perception and market positioning. By applying dimensionality reduction, we aim to transform 16 distinct performance and aesthetic features into a smaller set of interpretable "market pillars." These insights will support senior stakeholders in understanding brand strengths and identifying white-space opportunities in the competitive landscape.

Dataset Selection and Description

The analysis utilizes a dataset of 944 observations from the Brazilian automotive market (2013–2019). The data tracks various vehicle models (e.g., IX35, Duster, Compass) across 20 attributes.

- **Source:** Market research data focusing on SUV and crossover segments.
- **Key Attributes:** The dataset includes consumer-perceived scores for features such as brand, style, ruggedness, space, comfort, finish, infotainment, safety, and price. It also includes market performance metrics like share.
- **Relevance:** In a saturated market, understanding which features move together (e.g., do consumers associate "finish" with "safety"?) allows for more efficient product development and targeted marketing.

Analytical Approach and Model Development

To simplify the 16-dimensional feature space, Principal Component Analysis (PCA) was employed. The data was first standardized to ensure that variables with different scales (e.g., binary-like scores vs. continuous market share) did not disproportionately influence the model.

We evaluated three variations of the dimensionality reduction approach:

1. **Variation A: Full Feature PCA (16 Components)** This baseline model utilized all available features. While it captured 100% of the variance, it offered no simplification. However, it provided the "Scree Plot" necessary to determine the optimal number of factors.

2. **Variation B: Truncated PCA (Top 3 Components)** This model restricted the dimensions to the first three components, which together explained approximately 65% of the total variance. This offered high interpretability but potentially overlooked nuances related to secondary features like "economy" or "performance."

3. **Variation C: Optimized PCA (Top 5 Components)** This variation was selected as it reached the 80% cumulative variance threshold. It balanced the need for data simplification with the requirement to retain enough information to distinguish between "budget" and "premium" models effectively.

Model Selection and Justification

Variation C (5-Component PCA) is the recommended model for senior stakeholders.

- **Justification:** While Variation B was simpler, Variation C captures 80.3% of the market's complexity. In the automotive industry, where marginal differences in "infotainment" or "safety" can shift market share, retaining those extra dimensions is vital.

- **Interpretability:** The model successfully collapsed 16 variables into 5 intuitive "Market Pillars," making it easier for non-technical stakeholders to visualize where a specific model like the Jeep Compass sits compared to a Renault Duster.

Key Findings and Insights

The analysis revealed that the Brazilian automotive market is primarily defined by three dominant latent dimensions:

- **Pillar 1: Premium Sophistication (30.7% variance)** This is the strongest driver in the data. Features like safety, finish, style, and infotainment all move in tandem. Vehicles scoring high on this dimension are perceived as high-end, regardless of their mechanical "ruggedness."

- **Pillar 2: Utility vs. Agility (20.3% variance)** This dimension represents a trade-off. On one end sit vehicles focused on trunk space and roominess (utility); on the other end sit vehicles characterized by nimbleness and modern features (urban agility).

- **Pillar 3: The Efficiency/Versatility Axis (14.2% variance)** This dimension is heavily influenced by economy and versatility. It identifies a specific market segment that prioritizes low running costs and multi-purpose use over "brand status."

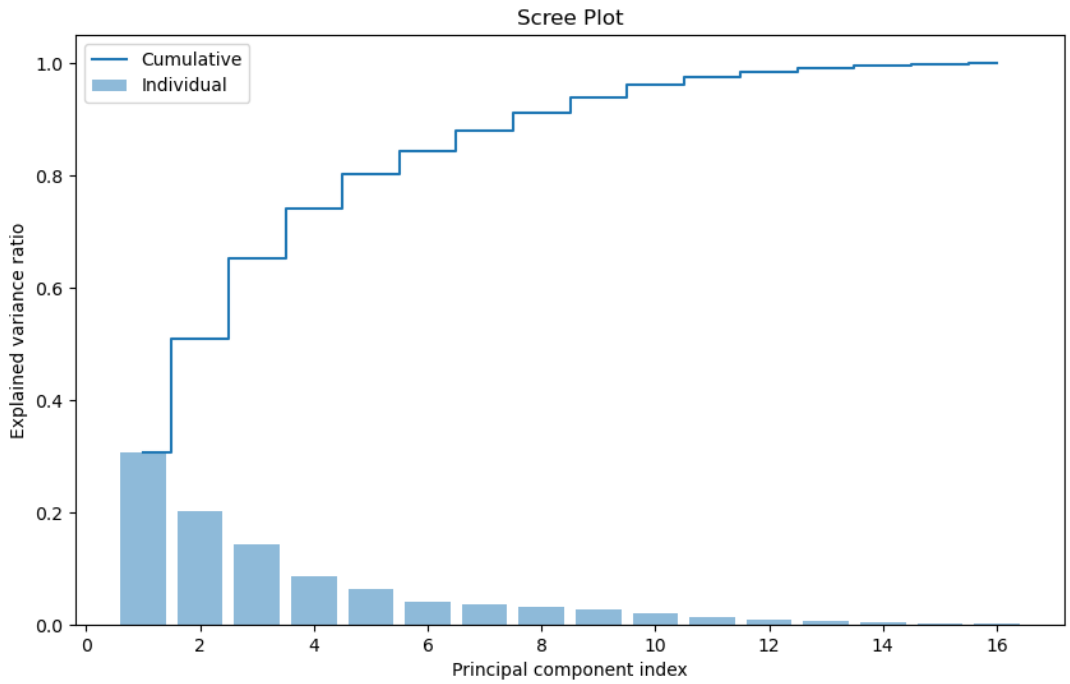
Managerial Implications: These findings suggest that "Safety" and "Style" are not distinct categories in the consumer's mind but are part of a single "Sophistication" pillar. Marketing campaigns should therefore bundle these attributes together. Furthermore, the inverse relationship between space and nimble in Pillar 2 suggests a clear market split between family-oriented SUVs and urban crossovers; trying to position a vehicle as "both" may confuse the target audience.

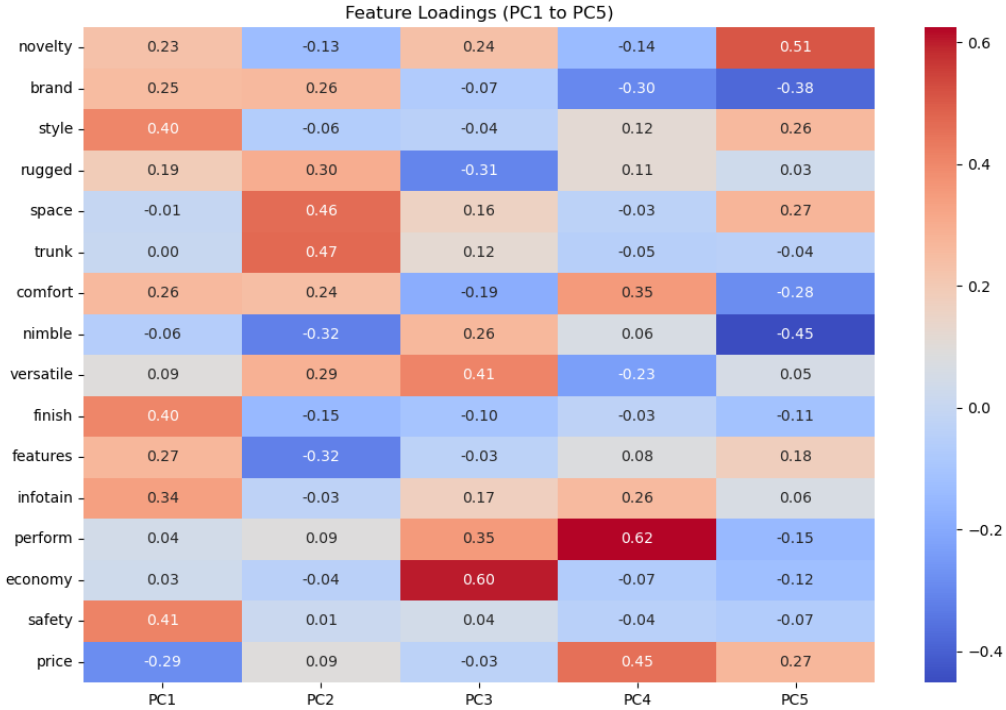
Limitations and Future Work

- **Data Limitations:** The dataset relies on subjective scoring (from consumer surveys), which can be prone to "halo effects" (e.g., a popular brand getting high scores in all categories regardless of actual performance).

- **Methodological Assumptions:** PCA assumes linear relationships between variables. If the relationship between price and share is non-linear (which is common in luxury goods), PCA might under-represent that impact.

- **Future Action Plan:** 1. **Clustering Integration:** The next phase should apply K-Means clustering to the 5 Principal Components identified here to segment the vehicles into specific "strategic groups." 2. **Temporal Analysis:** Incorporating the year and mo (month) data to see how these market pillars have shifted over time, especially during economic fluctuations in Brazil. 3. **Supervised Validation:** We propose using these 5 components as inputs for a regression model to predict share, testing if the simplified dimensions are better predictors than the 16 raw features.


Scree Plot
Cumulative
Individual
Explained variance ratio
Principal component index
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8
10
12
14
16


Feature Loadings (PC1 to PC5)
novelty 0.23 -0.13 0.24 -0.14 0.51
brand 0.25 0.26 -0.07 -0.30 -0.38
style 0.40 -0.06 -0.04 0.12 0.26
rugged 0.19 0.30 -0.31 0.11 0.03
space -0.01 0.46 0.16 -0.03 0.27
trunk 0.00 0.47 0.12 -0.05 -0.04
comfort 0.26 0.24 -0.19 0.35 -0.28
nimble -0.06 -0.32 0.26 0.06 -0.45
versatile 0.09 0.29 0.41 -0.23 0.05
finish 0.40 -0.15 -0.10 -0.03 -0.11
features 0.27 -0.32 -0.03 0.08 0.18
infotain 0.34 -0.03 0.17 0.26 0.06
perform 0.04 0.09 0.35 0.62 -0.15
economy 0.03 -0.04 0.60 -0.07 -0.12
safety 0.41 0.01 0.04 -0.04 -0.07
price -0.29 0.09 -0.03 0.45 0.27
PC1
PC2
PC3
PC4
PC5
0.6
0.4
0.2
0.0
−0.2
−0.4